MATH1025B
Essential Mathematical Methods

Chapter 8: Polynomials

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Definition 8.1

A **polynomial** $p(x)$ **with coefficients in** $\mathbb{R}$ is an infinite formal sum

$$\sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + \cdots + a_n x^n + \cdots$$

where $a_i \in \mathbb{R}$ and only finitely many of them are nonzero.

- If $a_i \neq 0$ for some $i \geq 0$, the largest such value of i is the **degree** of $p(x)$
- If $a_i = 0$ for all $i \geq 0$, then the degree of $p(x)$ is $-\infty$.

Remark:

(1) The a_i are **coefficients of** $p(x)$.

(2) The degree of $p(x)$ is denoted by $\deg(p(x))$.

(3) The set of all polynomials with coefficients in $\mathbb{R}$ is denoted by $\mathbb{R}[x]$.

(Although we primarily study $\mathbb{R}[x]$, all discussions work in $\mathbb{F}[x]$, where $\mathbb{F}$ is a field (e.g. $\mathbb{Q}$, $\mathbb{C}$).)

Polynomials

- To simplify working with polynomials, if
$$p(x) = \sum_{i=0}^{\infty} a_i x^i \text{ has } a_i = 0 \text{ for all } i > n,$$
we may denote $p(x) = a_0 + a_1 x + \cdots + a_n x^n$.
Caution: It does not mean $\deg(p(x)) = n$ unless we know $a_n \neq 0$.
- Furthermore, $p(x)$ is said to be **monic** if $a_n = 1$ (leading coefficient $= 1$).
- Write $1x^i$ in such a sum as x^i.
- Omit any term $0x^i$ unless all a_i are zero (i.e. when $p(x)$ is the zero polynomial).
- A number in $\mathbb{R}$ is a **constant polynomial**.

Definition 8.2

Let $p(x) = \sum_{i=0}^{\infty} a_i x^i$, $q(x) = \sum_{i=0}^{\infty} b_i x^i \in \mathbb{R}[x]$.

(1) (Sum) $p(x) + q(x)$ is defined as $\sum_{i=0}^{\infty} c_i x^i$, where $c_i = a_i + b_i$.

(2) (Product) $p(x)q(x)$ is defined as $\sum_{i=0}^{\infty} d_i x^i$, where $d_i = \sum_{k=0}^{i} a_k b_{i-k}$.

Observe that both c_i and d_i are 0 for all but a finite number of values of i, so these definitions make sense.

Proposition 8.1

Let $p(x), q(x) \in \mathbb{R}[x]$. Then

(1) $\deg(p(x) + q(x)) \leq \max\{\deg(p(x)), \deg(q(x))\}$.

(2) $\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x))$.

(Convention: $\max\{-\infty, -\infty\} = -\infty, \max\{-\infty, r\} = r$ and $(-\infty) + (-\infty) = (-\infty) + r = -\infty$ for any $r \geq 0$.)

Proof.
Suppose that $p(x), q(x) \in \mathbb{R}[x]$ are nonzero polynomials.
Let $p(x) = \sum_{i=0}^{\infty} a_k x^i$, $q(x) = \sum_{i=0}^{\infty} b_i x^i$ such that $\deg(p(x)) = r, \deg(q(x)) = s$.
Then, $a_r \neq 0$ and $a_i = 0$ for $i > r$; $b_s \neq 0$ and $b_i = 0$ for $i > s$.

(1) Note that $c_i = a_i + b_i = 0$ for $i > \max\{r, s\}$.
So $\deg(p(x) + q(x)) \leq \max\{r, s\}$.

(2) When $i > r + s$, i.e. $i - r \geq s + 1$, we have

$$d_i = \sum_{k=0}^{i} a_i b_{i-k} = a_0 b_i + a_1 b_{i-1} + \cdots + a_r b_{i-r} + a_{r+1} b_{i-r-1} + \cdots + a_i b_0 = 0.$$

Furthermore, $d_{r+s} = a_r b_s \neq 0$ ($\because a_r \neq 0, b_s \neq 0$).
So $\deg(p(x)q(x)) = r + s$.

For the case that $p(x)$ or $q(x)$ is zero, it is left as exercise. $\square$

Proposition 8.2 (Division Algorithm for polynomials)

Let $f(x), g(x) \in \mathbb{R}[x]$ such that $g(x)$ is nonzero, i.e. $\deg(g(x)) \geq 0$. Then, there exist unique $q(x), r(x) \in \mathbb{R}[x]$ such that

$$f(x) = g(x)q(x) + r(x), \qquad \textit{where } \deg(r(x)) < \deg(g(x)).$$

Proof of existence part.

Let $S = \{f(x) - g(x)s(x) : s(x) \in \mathbb{R}[x]\}$.

- If $0 \in S$, i.e. $f(x) = g(x)s_0(x)$ for some $s_0(x) \in \mathbb{R}[x]$.
  Take $q(x) = s_0(x)$, $r(x) = 0$. Done! $(\deg(r(x)) = -\infty < 0 \leq \deg(g(x)))$
- If $0 \notin S$, let $r_0(x) = f(x) - g(x)q_0(x) \in S$ be an element of minimal degree.
  Claim: $\deg(r_0(x)) < \deg(g(x))$.
  Suppose on the contrary that $\deg(r_0(x)) = k \geq j = \deg(g(x))$.
  Let

$$r_0(x) = c_0 + c_1 x + \cdots + c_k x^k \quad \text{with } c_k \neq 0 \text{ and}$$
$$g(x) = b_0 + b_1 x + \cdots + b_j x^j \quad \text{with } b_j \neq 0.$$

Division Algorithm

Consider

$$r_1(x) = r_0(x) - g(x)\left[(c_k/b_j)\,x^{k-j}\right],$$

which can be written in the form

$$r_1(x) = f(x) - g(x)\left[q_0(x) + (c_k/b_j)\,x^{k-j}\right].$$

So $r_1(x) \in S$.
Moreover,

$$\begin{aligned} r_1(x) &= \left(\sum_{i=0}^{k} c_i x^i\right) - \left(c_k x^k + \sum_{i=0}^{j-1} b_i \left(\frac{c_k}{b_j}\right) x^{i+k-j}\right) \\ &= \sum_{i=0}^{k-1} c_i x^i - \sum_{i=0}^{j-1} b_i \left(\frac{c_k}{b_j}\right) x^{i+k-j}, \end{aligned}$$

which is a polynomial with degree $< k$.
So $\deg(r_1(x)) < k \leq \deg(r_0(x))$, contradicting the fact that $r_0(x)$ was selected to have minimal degree in S.

Proof of uniqueness part.
Assume

$$f(x) = g(x)q_1(x) + r_1(x)$$

and

$$f(x) = g(x)q_2(x) + r_2(x)$$

for some $q_1(x), q_2(x), r_1(x), r_2(x) \in \mathbb{R}[x]$ with

$$\deg(r_1(x)), \deg(r_2(x)) < \deg(g(x)).$$

Then,

$$g(x)(q_1(x) - q_2(x)) = r_2(x) - r_1(x).$$

Now

$$\deg(g(x)(q_1(x) - q_2(x))) = \deg(g(x)) + \deg(q_1(x) - q_2(x))$$

while

$$\deg(r_2(x) - r_1(x)) \leq \max\{\deg(r_1(x)), \deg(r_2(x))\} < \deg(g(x)).$$

The equality can only hold if $LHS = RHS = -\infty$, i.e. $q_1(x) = q_2(x)$, $r_1(x) = r_2(x)$. $\square$

Example 8.0

Find the quotient and remainder in the division of $x^4 - 3x^3 + 2x^2 + 4x - 1$ by $x^2 - 2x + 3$.

By long division,

$$
\begin{array}{r|l}
 & \; x^2 - x - 3 \\
\hline
x^2 - 2x + 3 & x^4 - 3x^3 + 2x^2 + 4x - 1 \\
 & \underline{x^4 - 2x^3 + 3x^2} \\
 & \quad -x^3 - x^2 + 4x \\
 & \quad \underline{-x^3 + 2x^2 - 3x} \\
 & \qquad -3x^2 + 7x - 1 \\
 & \qquad \underline{-3x^2 + 6x - 9} \\
 & \qquad\qquad x + 8
\end{array}
$$

$$x^4 - 3x^3 + 2x^2 + 4x - 1 = (x^2 - 2x + 3)(x^2 - x - 3) + (x + 8),$$

$$\text{quotient} = x^2 - x - 3$$
$$\text{remainder} = x + 8.$$

Proposition 8.3 (Remainder Theorem)

Let $p(x) \in \mathbb{R}[x]$ and $a \in \mathbb{R}$. If $f(x)$ is divided by $(x-a)$, then the remainder is equal to $f(a)$.

Proof.
By division algorithm, there exist unique $q(x), r(x) \in \mathbb{R}[x]$ such that

$$f(x) = (x-a)q(x) + r(x)$$

and

$$\deg(r(x)) < \deg((x-a)) = 1.$$

Therefore $r(x) = R$ (a constant).
Put $x = a$ into $f(x) = (x-a)q(x) + R$, we have

$$\begin{aligned} f(a) &= (a-a)q(a) + R \\ R &= f(a). \end{aligned}$$

$\square$

Definition 8.3

Let $f(x), g(x) \in \mathbb{R}[x]$, we say that $g(x)$ **divides** $f(x)$ (or $f(x)$ is **divisible by** $g(x)$), denoted by $g(x) \mid f(x)$, if there exists $q(x) \in \mathbb{R}[x]$ such that $f(x) = g(x)q(x)$. In this case, $g(x)$ is a factor (or divisor) of $f(x)$.

Proposition 8.4 (Factor Theorem)

Let $f(x) \in \mathbb{R}[x]$ and $a \in \mathbb{R}$. Then $(x - a)$ is a factor of $f(x)$ if and only if $f(a) = 0$.

Proof.
By division algorithm, there exist unique $q(x) \in \mathbb{R}[x]$, $R \in \mathbb{R}$ such that

$$f(x) = (x - a)q(x) + R.$$

So $(x - a)$ is a factor of $f(x) \iff R = 0 \iff f(a) = 0$. $\square$

Example 8.1

Let $f(x) \in \mathbb{R}[x]$. Suppose that the remainders when $f(x)$ is divided by $(x-1)$ and $(x+1)$ are 5 and 3 respectively. Find the remainder when $f(x)$ is divided by $x^2 - 1 = (x-1)(x+1)$.

By the remainder theorem, $f(1) = 5$ and $f(-1) = 3$.
By division algorithm, there exists $q(x), ax + b \in \mathbb{R}[x]$ such that

$$f(x) = (x^2 - 1)q(x) + (ax + b).$$

Then

$$\begin{cases} f(1) = a + b & = 5 \\ f(-1) = -a + b & = 3 \end{cases} \implies \begin{cases} a = 1 \\ b = 4. \end{cases}$$

So the remainder is $x + 4$.

Corollary 8.5

A nonzero polynomial $f(x) \in \mathbb{R}[x]$ (resp. $\mathbb{C}[x]$) of degree n has at most n zeros in $\mathbb{R}$ (resp. $\mathbb{C}$).

Proof. We proceed by induction on n.
The case $n = 0$ is trivial because $f(x)$ is a nonzero constant, thus has no zeros.
Suppose any polynomial in $\mathbb{R}[x]$ of degree n has at most n zeros in $\mathbb{R}$.
Let $f(x) \in \mathbb{R}[x]$ of degree $n + 1$.
If $f(x)$ has no zero, then the result is trivial. So we assume that $f(x)$ has at least a zero $a \in \mathbb{R}$. By factor theorem, there exists $q(x) \in \mathbb{R}[x]$ such that

$$f(x) = (x - a)q(x).$$

Clearly $\deg(q(x)) = n$. By the induction hypothesis, $q(x)$ has at most n zeros in $\mathbb{R}$.
It is clear that any zero of $q(x)$ is also a zero of $f(x)$; and if $b \neq a$ is a zero of $f(x)$ then b is also a zero of $q(x)$.
Thus $f(x)$ has at most $n + 1$ zeros. $\square$

Corollary 8.6

Suppose that $p(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathbb{C}[x]$ *(i.e.* $\deg(p(x)) \leq n$*) has* $n+1$ *distinct zeros in* $\mathbb{C}$*. Then* $p(x)$ *is the zero polynomial.*

Exercise 8.1

Let $f(x) \equiv a_0 + a_1 x + \cdots + a_m x^m$ and $g(x) \equiv b_0 + b_1 x + \cdots + b_n x^n$ be polynomials with real coefficients. Define

$$F(x) \equiv f(x^3) \text{ and } G(x) \equiv g(x^3).$$

Suppose $F(x) + xG(x)$ is divisible by $x^2 + x + 1$, prove that both $f(x)$ and $g(x)$ are divisible by $x - 1$.

Hint: Note that $x^2 + x + 1 = (x - \omega)(x - \omega^2)$ where $\omega = e^{2\pi i/3}$.

Division Algorithm

Proposition (Division Algorithm for polynomials)

Let $f(x), g(x) \in \mathbb{R}[x]$ *such that* $g(x)$ *is nonzero, i.e.* $\deg(g(x)) \geq 0$. *Then, there exist unique* $q(x), r(x) \in \mathbb{R}[x]$ *such that*

$$f(x) = g(x)q(x) + r(x), \qquad \textit{where } \deg(r(x)) < \deg(g(x)).$$

Proposition (Remainder Theorem)

Let $p(x) \in \mathbb{R}[x]$ *and* $a \in \mathbb{R}$. *If* $f(x)$ *is divided by* $(x - a)$, *then the remainder is equal to* $f(a)$.

Proposition (Factor Theorem)

Let $f(x) \in \mathbb{R}[x]$ *and* $a \in \mathbb{R}$. *Then* $(x - a)$ *is a factor of* $f(x)$ *if and only if* $f(a) = 0$.

Corollary

Suppose that $p(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathbb{C}[x]$ *(i.e.* $\deg(p(x)) \leq n$*) has* $n + 1$ *distinct zeros in* $\mathbb{C}$. *Then* $p(x)$ *is the zero polynomial.*

Lemma 8.7

Let $f(x), g(x) \in \mathbb{R}[x]$ such that $f(x), g(x)$ are nonzero. Then $g(x) \mid f(x)$ and $f(x) \mid g(x)$ if and only if $f(x) = kg(x)$ for some nonzero $k \in \mathbb{R}$.

Proof.
($\Longleftarrow$) Trivial! ($\because f(x) = kg(x)$ and $g(x) = \frac{1}{k}f(x)$.)
($\Longrightarrow$) Assume $g(x) \mid f(x)$ and $f(x) \mid g(x)$.
Then, $f(x) = d_1(x)g(x)$ and $g(x) = d_2(x)f(x)$ for some $d_1(x), d_2(x) \in \mathbb{R}[x]$.
Now,

$$\begin{aligned} & f(x) = d_1(x)d_2(x)f(x) \\ \Longrightarrow \quad & d_1(x)d_2(x) = 1 \quad \text{(by } f(x) \neq 0 \text{ and division algorithm)} \\ \Longrightarrow \quad & \text{both } d_1(x) \text{ and } d_2(x) \text{ are constant polynomials.} \end{aligned}$$

So we can write

$$d_1(x) = k \text{ and } d_2(x) = \frac{1}{k}, \qquad \text{for some } k \in \mathbb{R}\backslash\{0\}. \qquad \square$$

GCD for Polynomials

Recall the GCD of two integers.

Definition

Let $a, b \in \mathbb{Z}$. The **greatest common divisor** of a and b is defined as

$$\gcd(a,b) = \begin{cases} \max\{d \in \mathbb{Z} : d \mid a \text{ and } d \mid b\} & \text{if } a \neq 0 \text{ or } b \neq 0 \\ 0 & \text{if } a = b = 0. \end{cases}$$

(Therefore, if $a \neq 0$, $\gcd(a, 0) = |a|$.)

Alternative definition of greatest common divisor

Let $a, b \in \mathbb{Z}$ and let $d \in \mathbb{N}$. The statements (1), (2) are equivalent:

(1) $d = \gcd(a, b)$

(2) *d is a common divisor of a, b, and d is divisible by every common divisor of a, b.*

GCD for Polynomials

Definition 8.4

Let $f(x), g(x) \in \mathbb{R}[x]$ that are not both zero. A polynomial $d(x) \in \mathbb{R}[x]$ is said to be a **greatest common divisor** of $f(x)$ and $g(x)$ if

(1) $d(x)$ is a common factor of $f(x)$ and $g(x)$,
i.e. $d(x) \mid f(x)$ and $d(x) \mid g(x)$; and

(2) if $d'(x)$ is a common factor of $f(x)$ and $g(x)$, then $d'(x) \mid d(x)$.

- If $d_1(x), d_2(x)$ are greatest common divisors of $f(x)$ and $g(x)$, then, $d_1(x) \mid d_2(x)$ and $d_2(x) \mid d_1(x)$. So, by Lemma 8.7,

$$d_1(x) = kd_2(x) \quad \text{for some nonzero } k \in \mathbb{R}.$$

- Greatest common divisor of $f(x)$ and $g(x)$ is unique up to the multiplication by a nonzero constant.
- However, it is unique if we take a monic polynomial (i.e. the leading coefficient is 1). We denote it by $\gcd(f(x), g(x))$.

GCD for Polynomials

Let $f(x), g(x)$ be nonzero polynomials in $\mathbb{R}[x]$. Then, there exist unique $q(x), r(x) \in \mathbb{R}[x]$ such that

$$f(x) = g(x)q(x) + r(x), \quad \text{where } \deg(r(x)) < \deg(g(x)).$$

Proposition 8.8

Then $\gcd(f(x), g(x)) = \gcd(g(x), r(x))$

Proof.
Let $d(x) = \gcd(f(x), g(x))$ and $d'(x) = \gcd(g(x), r(x))$. Then

$$\begin{aligned} d'(x) \mid g(x) \text{ and } d'(x) \mid r(x) &\implies d'(x) \mid f(x) = g(x)q(x) + r(x) \text{ and } d'(x) \mid g(x) \\ &\implies d'(x) \text{ is a common factor of } f(x), g(x) \\ &\implies d'(x) \mid d(x), \end{aligned}$$

and

$$\begin{aligned} d(x) \mid f(x) \text{ and } d(x) \mid g(x) &\implies d(x) \mid r(x) = f(x) - g(x)q(x) \text{ and } d(x) \mid g(x) \\ &\implies d(x) \text{ is a common factor of } g(x), r(x) \\ &\implies d(x) \mid d'(x), \end{aligned}$$

Since $d(x), d'(x)$ are both monic, we have $d(x) = d'(x)$. $\square$

Proposition 8.9 (Extended Euclidean Algorithm)

Let $f(x), g(x)$ be nonzero polynomials in $\mathbb{R}[x]$.
Then, there exist $a(x), b(x) \in \mathbb{R}[x]$ such that

$$\gcd(f(x), g(x)) = a(x)f(x) + b(x)g(x).$$

Proof.
Repeat the proof of extended Euclidean algorithm for integers. □
Note that we have a sequence of remainders $r_k(x)$ such that

$$\deg(g(x)) > \deg(r_2(x)) > \cdots > \deg(r_{n-1}(x)) > \deg(r_n(x)).$$

Definition 8.5

Let $f(x), g(x)$ be nonzero polynomials in $\mathbb{R}[x]$.
Then $f(x)$ and $g(x)$ are said to be **relatively prime** if $\gcd(f(x), g(x)) = 1$.
(Equivalently, there exist $a(x), b(x) \in \mathbb{R}[x]$ such that
$a(x)f(x) + b(x)g(x) = 1$.)

Example 8.2

Let $f(x) = 4x^4 - 2x^3 - 16x^2 + 5x + 9$, $g(x) = 2x^3 - x^2 - 5x + 4 \in \mathbb{R}[x]$. Find $\gcd(f(x), g(x))$ and $a(x), b(x) \in \mathbb{R}[x]$ such that

$$\gcd(f(x), g(x)) = a(x)f(x) + b(x)g(x).$$

By Euclidean algorithm,

$$\begin{aligned}
4x^4 - 2x^3 - 16x^2 + 5x + 9 &= (2x)(2x^3 - x^2 - 5x + 4) + (-6x^2 - 3x + 9) \\
2x^3 - x^2 - 5x + 4 &= (-\frac{1}{3}x + \frac{1}{3})(-6x^2 - 3x + 9) + (-x + 1) \\
-6x^2 - 3x + 9 &= (6x + 9)(-x + 1) + 0.
\end{aligned}$$

Hence $\gcd(f(x), g(x)) = x - 1$. Moreover,

$$\begin{aligned}
x - 1 &= (-\frac{1}{3}x + \frac{1}{3})(-6x^2 - 3x + 9) - g(x) \\
&= (-\frac{1}{3}x + \frac{1}{3})[f(x) - (2x)g(x)] - g(x) \\
&= (-\frac{1}{3}x + \frac{1}{3})f(x) + (\frac{2}{3}x^2 - \frac{2}{3}x - 1)g(x).
\end{aligned}$$

Irreducible Polynomials

Definition 8.6

A nonconstant $f(x) \in \mathbb{F}[x]$ ($\mathbb{F} = \mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$) is **irreducible over** $\mathbb{F}$ if $f(x)$ cannot be expressed as a product of $g(x), h(x) \in \mathbb{F}[x]$ with $\deg(g(x)), \deg(h(x)) < \deg(f(x))$.

(Analogous to prime in $\mathbb{Z}$.)

Example 8.3

- Let $x^2 - 2 \in \mathbb{Q}[x]$. Then it is irreducible over $\mathbb{Q}$.
  However, if we consider $x^2 - 2 \in \mathbb{R}[x]$, then it is reducible over $\mathbb{R}$ since
  $$x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2}).$$
- Let $x^2 + 1 \in \mathbb{R}[x]$. Then it is irreducible over $\mathbb{R}$.
  However, if we consider $x^2 + 1 \in \mathbb{C}[x]$, then it is reducible over $\mathbb{C}$ since
  $$x^2 + 1 = (x - i)(x + i).$$

Irreducible Polynomials

By the fundamental theorem of algebra, a degree n polynomial $p(x) \in \mathbb{C}[x]$ can be factorized as

$$p(x) = a_n(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n),$$

where $\alpha_i \in \mathbb{C}$ are the not necessarily distinct roots of $p(x)$.
Therefore,

(1) All irreducible polynomials in $\mathbb{C}[x]$ are those linear (degree 1) polynomials.

(2) All irreducible polynomials in $\mathbb{R}[x]$ are those linear (degree 1) polynomials and quadratic (degree 2) polynomials in the form $ax^2 + bx + c$ with $\Delta = b^2 - 4ac < 0$.
(Since $p(\alpha) = 0 \iff p(\overline{\alpha}) = 0$, and when $\alpha \notin \mathbb{R}$,

$$(x - \alpha)(x - \overline{\alpha}) = x^2 - 2\,\mathrm{Re}(\alpha)x + |\alpha|^2 \in \mathbb{R}[x] \text{ and is irreducible.)}$$

Irreducible Polynomials

Proposition 8.10

Let $a(x), p(x) \in \mathbb{R}[x]$ *such that* $p(x)$ *is irreducible over* $\mathbb{R}$. *If* $p(x) \nmid a(x)$, *then* $\gcd(a(x), p(x)) = 1$.

Proposition 8.11

Let $p(x), a(x), b(x) \in \mathbb{R}[x]$. *If* $p(x)$ *is irreducible over* $\mathbb{R}$ *and* $p(x) \mid a(x)b(x)$, *then either* $p(x) \mid a(x)$ *or* $p(x) \mid b(x)$.

Proposition 8.12

Let $a(x), b(x), c(x) \in \mathbb{R}[x]$. *Then there exist* $p(x), q(x) \in \mathbb{R}[x]$ *such that* $a(x)p(x) + b(x)q(x) = c(x)$ *if and only if* $\gcd(a(x), b(x)) \mid c(x)$.

Proposition 8.13

Every nonconstant $f(x) \in \mathbb{F}[x]$ *can be factored into a product of irreducible polynomials in* $\mathbb{F}[x]$, *the irreducible polynomial being unique except for order and for nonzero constant factors.*

Exercise 8.2

Prove the propositions above.

(**Hint:** Refer to Exercise 7.2, Proposition 7.14, 7.19 and Theorem 7.16.)

Rational Functions and Partial Fraction Decomposition

All polynomials in this part are real polynomials.

Definition 8.7

A **rational function** is a quotient $\dfrac{p(x)}{q(x)}$ of two polynomials $p(x)$ and $q(x)$ where $q(x)$ is nonzero.

In calculus, we are sometimes asked to evaluate integrals of rational functions, for example:

$$\int \frac{2x^2+1}{x^2(x^2+1)^2}\,dx.$$

Goal

Express a rational function $\dfrac{p(x)}{q(x)}$ as a sum of polynomial and simpler fractions **(partial fraction decomposition)**.

Proposition 8.14 (Partial Fraction Decomposition)

Let $p(x), q(x) \in \mathbb{R}[x]$ such that $q(x)$ can be uniquely factorized as a product in form of

$$K \cdot \prod_{i=1}^{r} (x - a_i)^{m_i} \cdot \prod_{j=1}^{s} (x^2 + b_j x + c_j)^{n_j}$$

where $K, a_i, b_i, c_i \in \mathbb{R}$, $m_i, n_j \in \mathbb{Z}^+$, $\deg(q(x)) = (\sum_{i=1}^{r} m_i) + 2(\sum_{j=1} n_j)$. Then the rational function $\dfrac{p(x)}{q(x)}$ can be expressed uniquely in form of

$$\frac{p(x)}{q(x)} = b(x) + \frac{1}{K} \left(\sum_{i=1}^{r} \sum_{\alpha=1}^{m_i} \frac{k_{i_\alpha}}{(x - a_i)^\alpha} + \sum_{j=1}^{s} \sum_{\beta}^{n_j} \frac{r_{j_\beta} x + s_{j_\beta}}{(x^2 + b_j x + c_j)^\beta} \right),$$

where $b(x) \in \mathbb{R}[x]$ and $k_{i_\alpha}, r_{j_\beta}, s_{j_\beta} \in \mathbb{R}$.

(1) Polynomial Part

If $\deg(p(x)) \geq \deg(q(x))$, by the division algorithm, there exist unique $b(x), r(x) \in \mathbb{R}[x]$ such that

$$p(x) = q(x)b(x) + r(x) \quad \text{and} \quad \deg(r(x)) < \deg(q(x)).$$

Therefore,

$$\frac{p(x)}{q(x)} = b(x) + \frac{r(x)}{q(x)}.$$

So we focus on $\dfrac{r(x)}{q(x)}$ with $\deg(r(x)) < \deg(q(x))$.

(2) Factors of the Denominator

Consider $\dfrac{r(x)}{q(x)}$ with $\deg(r(x)) < \deg(q(x))$. Suppose $q(x) = q_1(x)q_2(x)$ for some relatively prime polynomials $q_1(x)$ and $q_2(x)$.

Claim: There exist unique $r_1(x), r_2(x) \in \mathbb{R}[x]$ with $\deg(r_1(x)) < \deg(q_1(x)), \deg(r_2(x)) < \deg(q_2(x))$ such that

$$\frac{r(x)}{q(x)} = \frac{r_1(x)}{q_1(x)} + \frac{r_2(x)}{q_2(x)}.$$

Existence: Since $\gcd(q_1(x), q_2(x)) = 1$, there exist $a(x), b(x)$ such that $a(x)q_1(x) + b(x)q_2(x) = 1$.

$$\begin{aligned}\frac{r(x)}{q(x)} &= \frac{r(x)[a(x)q_1(x) + b(x)q_2(x)]}{q_1(x)q_2(x)} \\ &= \frac{r(x)b(x)}{q_1(x)} + \frac{r(x)a(x)}{q_2(x)} \\ &= c_1(x) + c_2(x) + \frac{r_1(x)}{q_1(x)} + \frac{r_2(x)}{q_2(x)}\end{aligned}$$

Division algorithm:

$$\begin{cases} r(x)b(x) = c_1(x)q_1(x) + r_1(x) \\ r(x)a(x) = c_2(x)q_1(x) + r_2(x) \end{cases}$$

$\deg(r_i(x)) < \deg(q_i(x))$, $i = 1, 2$

Since $\deg(r(x)) < \deg(q(x))$, we have $c_1(x) + c_2(x) = 0$.

Uniqueness: Suppose that

$$\frac{r(x)}{q(x)} = \frac{r_1(x)}{q_1(x)} + \frac{r_2(x)}{q_2(x)} = \frac{\tilde{r}_1(x)}{q_1(x)} + \frac{\tilde{r}_2(x)}{q_2(x)}$$

with

$$\deg(r_i(x)), \deg(\tilde{r}_i(x)) < \deg(q_i(x)) \ , \ i = 1, 2.$$

Then,

$$\begin{aligned}
& (r_1(x) - \tilde{r}_1(x))q_2(x) = (r_2(x) - \tilde{r}_2(x))q_1(x) \\
\implies \quad & q_1(x) \mid (r_1(x) - \tilde{r}_1(x))q_2(x) \\
\implies \quad & q_1(x) \mid (r_1(x) - \tilde{r}_1(x)) \qquad (\because q_1(x), q_2(x) \text{ are relatively prime}) \\
\implies \quad & r_1(x) - \tilde{r}_1(x) = 0 \qquad (\because \deg(r_1(x) - \tilde{r}_1(x)) < \deg(q_1(x)))
\end{aligned}$$

Hence $r_1(x) = \tilde{r}_1(x)$. Similarly, $r_2(x) = \tilde{r}_2(x)$.

Recall: A real polynomial $q(x)$ can be uniquely factorized as a product in form of

$$K \cdot \prod_{i=1}^{r} (x - a_i)^{m_i} \cdot \prod_{j=1}^{s} (x^2 + b_j x + c_j)^{n_j}$$

where $K, a_i, b_i, c_i \in \mathbb{R}$, $m_i, n_j \in \mathbb{Z}^+$, $\deg(q(x)) = \left(\sum_{i=1}^{r} m_i\right) + 2\left(\sum_{j=1} n_j\right)$.
Moreover, any pair of linear or quadratic factors are relatively prime.
Therefore, $\dfrac{r(x)}{q(x)}$ with $\deg(r(x)) < \deg(q(x))$ can be expressed uniquely in form of

$$\frac{r(x)}{q(x)} = \frac{1}{K}\left(\sum_{i=1}^{r} \frac{A_i(x)}{(x - a_i)^{m_i}} + \sum_{j=1}^{s} \frac{B_j(x)}{(x^2 + b_j x + c_j)^{n_j}}\right)$$

where $\deg(A_i(x)) < m_i$ and $\deg(B_j(x)) < 2n_j$.
So we focus on

$$\frac{A_i(x)}{(x - a_i)^{m_i}} \quad \text{and} \quad \frac{B_j(x)}{(x^2 + b_j x + c_j)^{n_j}}.$$

(3) Powers in the Denominator

Case 1: $\dfrac{A(x)}{(x-a)^m}$ where $\deg(A(x)) < m$.

When $m > 1$, by the division algorithm, there exist unique $\tilde{A}(x)$, k_m such that $A(x) = (x-a)\tilde{A}(x) + k_m$ and so $\dfrac{A(x)}{(x-a)^m} = \dfrac{k_m}{(x-a)^m} + \dfrac{\tilde{A}(x)}{(x-a)^{m-1}}$.

Repeating the above until $m = 1$.

So $\dfrac{A(x)}{(x-a)^m}$ can be expressed uniquely in form of $\displaystyle\sum_{i=1}^{m} \frac{k_i}{(x-a)^i}$.

Case 2: $\dfrac{B(x)}{(x^2+cx+b)^n}$ where $\deg(B(x)) < 2n$.

When $n > 1$, by the division algorithm, there exist unique $\tilde{B}(x)$, $r_x + s_n$ such that $B(x) = (x^2+bx+c)\tilde{B}(x) + (r_n x + s_n)$ and so

$$\frac{B(x)}{(x^2+cx+b)^n} = \frac{r_n x + s_n}{(x^2+bx+c)^n} + \frac{\tilde{B}(x)}{(x^2+bx+c)^{n-1}}.$$

Repeating the above until $n = 1$.

So $\dfrac{B(x)}{(x^2+cx+b)^n}$ can be expressed uniquely in form of $\displaystyle\sum_{j=1}^{n} \frac{r_j x + s_j}{(x^2+bx+c)^j}$.

Example 8.4

Resolve $\dfrac{x^4+2x+4}{2x^3-4x^2+3x-6}$ into partial fractions.

By long division,

$$
\begin{array}{r|l}
 & \tfrac{1}{2}x+1 \\
\hline
2x^3-4x^2+3x-6 & x^4+0x^3+0x^2+2x+4 \\
 & \underline{x^4-2x^3+\tfrac{3}{2}x^2-3x} \\
 & 2x^3-\tfrac{3}{2}x^2+5x+4 \\
 & \underline{2x^3-4x^2+3x-6} \\
 & \tfrac{5}{2}x^2+2x+10
\end{array}
$$

So

$$
\begin{aligned}
\frac{x^4+2x+4}{2x^3-4x^2+3x-6} &= \frac{1}{2}x+1+\frac{\frac{5}{2}x^2+2x+10}{2x^3-4x^2+3x-6} \\
&= \frac{1}{2}x+1+\frac{\frac{5}{2}x^2+2x+10}{(2x^2+3)(x-2)}.
\end{aligned}
$$

Example 8.4

Resolve $\dfrac{x^4+2x+4}{2x^3-4x^2+3x-6}$ into partial fractions.

Write

$$\frac{\frac{5}{2}x^2+2x+10}{(2x^2+3)(x-2)} = \frac{Ax+B}{2x^2+3} + \frac{C}{x-2}.$$

So

$$\begin{aligned}\frac{5}{2}x^2+2x+10 &= (Ax+B)(x-2)+C(2x^2+3)\\ &= (A+2C)x^2+(-2A+B)x+(-2B+3C)\end{aligned}$$

$$\begin{cases} A+2C=\frac{5}{2} \\ -2A+B=2 \\ -2B+3C=10 \end{cases} \implies \begin{array}{c} A=-\frac{41}{22} \\ B=-\frac{19}{11} \\ C=\frac{24}{11} \end{array}$$

Hence

$$\frac{x^4+2x+4}{2x^3-4x^2+3x-6} = \frac{1}{2}x+1-\frac{41x+38}{22(2x^2+3)}+\frac{24}{11(x-2)}.$$

Example 8.5

Write down the forms when the following rational functions are resolved into partial fractions.

(a) $\dfrac{5}{x^2+x-6}$

(b) $\dfrac{x^3+1}{(x-2)^4}$

(c) $\dfrac{2x^2+1}{x^2(x^2+1)^2}$

(a) $$\frac{5}{x^2+x-6} = \frac{A}{x+3} + \frac{B}{x-2} \qquad (A=-1, B=1)$$

(b) $$\frac{x^3+1}{(x-2)^4} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4}$$
$$(A=1, B=6, C=12, D=9)$$

(c) $$\frac{2x^2+1}{x^2(x^2+1)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} + \frac{Ex+F}{(x^2+1)^2}$$
$$(A=0, B=1, C=0 D=-1, E=0, F=1)$$

Example 8.6

Resolve $\dfrac{x}{(2x-1)(2x+1)(2x+3)}$ into partial fractions.

Hence find $\displaystyle\sum_{k=1}^{n} \frac{k}{(2k-1)(2k+1)(2k+3)}$.

Let $\dfrac{x}{(2x-1)(2x+1)(2x+3)} = \dfrac{A}{2x-1} + \dfrac{B}{2x+1} + \dfrac{C}{2x+3}$.

Then $A = \dfrac{1}{16}, B = \dfrac{1}{8}, C = -\dfrac{3}{16}$ (Exercise). Now,

$$\sum_{k=1}^{n} \frac{k}{(2k-1)(2k+1)(2k+3)} = \frac{1}{16}\left(\sum_{k=1}^{n} \frac{1}{2k-1} + \sum_{k=1}^{n} \frac{2}{2k+1} - \sum_{k=1}^{n} \frac{3}{2k+3}\right)$$

$$= \frac{1}{16}\left[\left(\frac{1}{1} + \frac{1}{3} + \underbrace{\sum_{k=3}^{n} \frac{1}{2k-1}}_{=\sum_{k=1}^{n-2} \frac{1}{2k+3}}\right) + \left(\frac{2}{3} + \underbrace{\sum_{k=2}^{n-1} \frac{2}{2k+1}}_{=\sum_{k=1}^{n-2} \frac{2}{2k+3}} + \frac{2}{2n+1}\right) - \left(\underbrace{\sum_{k=1}^{n-2} \frac{3}{2k+3}}_{=\sum_{k=1}^{n-2} \frac{3}{2k+3}} + \frac{3}{2n+1} + \frac{3}{2n+3}\right)\right]$$

$$= \frac{1}{16}\left[2 - \frac{1}{2n+1} - \frac{3}{2n+3}\right].$$

Relationship between Roots and Coefficients of Polynomials

Suppose $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{C}[x]$ where $a_n \neq 0$.
By the fundamental theorem of algebra,

$$p(x) = a_n(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n)$$

for some $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{C}$ (not necessarily distinct).
Then, we expand the above expression:

$$\begin{aligned} p(x) = a_n[x^n &- (\alpha_1 + \alpha_2 + \cdots + \alpha_n)x^{n-1} + \\ &(\alpha_1\alpha_2 + \alpha_1\alpha_3 + \cdots + \alpha_{n-1}\alpha_n)x^{n-2} - \cdots + (-1)^n(\alpha_1\alpha_2\cdots\alpha_n)]. \end{aligned}$$

Comparing coefficients:

$$\begin{cases} \alpha_1 + \alpha_2 + \cdots + \alpha_n &= -\dfrac{a_{n-1}}{a_n} \\ \alpha_1\alpha_2 + \alpha_1\alpha_3 + \cdots + \alpha_{n-1}\alpha_n &= \dfrac{a_{n-2}}{a_n} \\ \vdots & \\ \alpha_1\alpha_2\cdots\alpha_n &= (-1)^n\dfrac{a_0}{a_n} \end{cases}$$

Relationship between Roots and Coefficients of Polynomials

Theorem 8.15 (Vieta's Formulas)

Suppose $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{C}[x]$, *where* $a_n \neq 0$, *has* n *(not necessarily distinct) complex roots* $\alpha_1, \alpha_2, \ldots, \alpha_n$. *Then*

$$\sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} \left(\prod_{k=1}^{j} \alpha_{i_k} \right) = (-1)^j \frac{a_{n-j}}{a_n}, \qquad \text{for } j = 1, 2, \ldots, n.$$

The indices i_k are sorted in increasing order to ensure each product of j roots is used exactly once.

Equivalently, Vieta's formulas can be expressed as follows:
Let $I \subseteq \{1, 2, \ldots, n\}$, write $\alpha_I = \alpha_{i_1}\alpha_{i_2}\cdots\alpha_{i_j}$ where $i_1, i_2, \ldots, i_j$ are all elements in I. Then

$$\sum_{\substack{I \subseteq \{1,2,\ldots,n\} \\ |I| = j}} \alpha_I = (-1)^j \frac{a_{n-j}}{a_n}.$$

Example 8.7

Let $\alpha_1, \alpha_2, \alpha_2 \in \mathbb{C}$ be the roots of $a_0 + a_1x + a_2x^2 + a_3x^3 = 0$ and $S_r = \alpha_1^r + \alpha_2^r + \alpha_3^r$ for $r = 1, 2, \ldots$.
Express S_2 and S_3 in terms of a_i's.

Firstly,

$$\begin{cases} \alpha_1 + \alpha_2 + \alpha_3 &= -\dfrac{a_2}{a_3} \\ \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3 &= \dfrac{a_1}{a_3} \\ \alpha_1\alpha_2\alpha_3 &= -\dfrac{a_0}{a_3} \end{cases}.$$

Then,

$$\begin{aligned} S_2 &= \alpha_1^2 + \alpha_2^2 + \alpha_3^2 \\ &= (\alpha_1 + \alpha_2 + \alpha_3)^2 - 2(\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3) \\ &= \frac{a_2^2 - 2a_1a_3}{a_3^2}. \end{aligned}$$

Example 8.7

Let $\alpha_1, \alpha_2, \alpha_2 \in \mathbb{C}$ be the roots of $a_0 + a_1x + a_2x^2 + a_3x^3 = 0$ and $S_r = \alpha_1^r + \alpha_2^r + \alpha_3^r$ for $r = 1, 2, \ldots$.
Express S_2 and S_3 in terms of a_i's.

$$\begin{aligned}
S_3 &= \alpha_1^3 + \alpha_2^3 + \alpha_3^3 \\
&= (\alpha_1 + \alpha_2 + \alpha_3)^3 - 3(\alpha_1^2\alpha_2 + \alpha_1^2\alpha_3 + \alpha_2^2\alpha_1 + \alpha_2^2\alpha_3 + \alpha_3^2\alpha_1 + \alpha_3^2\alpha_2) \\
&\quad - 6\alpha_1\alpha_2\alpha_3 \\
&= (\alpha_1 + \alpha_2 + \alpha_3)^3 - 3\left[(\alpha_1^2 + \alpha_2^2 + \alpha_3^2)(\alpha_1 + \alpha_2 + \alpha_3) - (\alpha_1^3 + \alpha_2^3 + \alpha_3^3)\right] \\
&\quad - 6\alpha_1\alpha_2\alpha_3.
\end{aligned}$$

So

$$\begin{aligned}
S_3 &= \left(-\frac{a_2}{a_3}\right)^3 - 3\left[\left(\frac{a_2^2 - 2a_1a_3}{a_3^2}\right)\left(-\frac{a_2}{a_3}\right) - S_3\right] - 6\left(-\frac{a_0}{a_3}\right) \\
-2S_3 &= \frac{1}{a_3^3}\left(-a_2^3 + 3a_2^3 - 6a_1a_2a_3 + 6a_0a_3^2\right) \\
S_3 &= \frac{1}{a_3^3}\left(-a_2^3 + 3a_1a_2a_3 - 3a_0a_3^2\right).
\end{aligned}$$

Lagrange Interpolation (Optional)

Let $(x_1, y_1), (x_2, y_2), \ldots, (x_{n+1}, y_{n+1}) \in \mathbb{R}^2$ such that all x_i's are distinct.

Question

Can we find a polynomial $f(x)$ of degree n (or less) such that $f(x_i) = y_i$ for all $i = 1, 2, \ldots, n+1$?

Let

$$f_i(x) = \frac{\prod_{1 \le j \le n+1, j \ne i}(x - x_j)}{\prod_{1 \le j \le n+1, j \ne i}(x_i - x_j)} \qquad (\text{for } i = 1, 2, \ldots, n+1)$$

which is a polynomial of degree n that satisfies

$$f_i(x_j) = \delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \ne j. \end{cases}$$

Then $f(x) := \sum_{i=1}^{n+1} y_i f_i(x)$ is a polynomial of degree $\le n$ such that

$$f(x_j) = \sum_{i=1}^{n} y_i f_i(x_j) = \sum_{i=1}^{n} y_i \delta_{ij} = y_j \qquad \text{for } j = 1, 2, \ldots, n+1.$$

Such a polynomial is unique because the difference of any two such polynomials is a polynomial of degree $\le n$ with $n+1$ zeros.

Exercise 8.3

(a) Show, using de Moivre's theorem, or otherwise, that

$$\tan 9\theta = \frac{t(t^2-3)(t^6-33t^4+27t^2-3)}{(3t^2-1)(3t^6-27t^4+33t^2-1)}, \qquad \text{where } t = \tan\theta.$$

(b) By considering the equation $\tan 9\theta = 0$, or otherwise, obtain a cubic equation with integer coefficients whose roots are

$$\tan^2\left(\frac{\pi}{9}\right), \qquad \tan^2\left(\frac{2\pi}{9}\right) \qquad \text{and} \qquad \tan^2\left(\frac{4\pi}{9}\right).$$

(c) Deduce the value of

$$\tan\left(\frac{\pi}{9}\right)\tan\left(\frac{2\pi}{9}\right)\tan\left(\frac{4\pi}{9}\right).$$

(d) Show that

$$\tan^6\left(\frac{\pi}{9}\right) + \tan^6\left(\frac{2\pi}{9}\right) + \tan^6\left(\frac{4\pi}{9}\right) = 33273.$$

Exercise 8.3

(a) Show, using de Moivre's theorem, or otherwise, that

$$\tan 9\theta = \frac{t(t^2-3)(t^6-33t^4+27t^2-3)}{(3t^2-1)(3t^6-27t^4+33t^2-1)}, \qquad \text{where } t = \tan\theta.$$

(a) By de Moivre's theorem,

$$\begin{aligned}
\tan 9\theta &= \frac{(\cos 9\theta + i\sin 9\theta) - (\cos 9\theta - i\sin 9\theta)}{(\cos 9\theta + i\sin 9\theta) + (\cos 9\theta - i\sin 9\theta)} \cdot \frac{1}{i} \\
&= \frac{(\cos\theta + i\sin\theta)^9 - (\cos\theta - i\sin\theta)^9}{(\cos\theta + i\sin\theta)^9 + (\cos\theta - i\sin\theta)^9} \cdot \frac{1}{i} \\
&= \frac{(1+it)^9 - (1-it)^9}{(1+it)^9 + (1-it)^9} \cdot \frac{1}{i}.
\end{aligned}$$

The formula now follows since

$$\begin{aligned}
(1+it)^9 - (1-it)^9 &= 2it(t^8 - 36t^6 + 126t^4 - 84t^2 + 9) \\
&= 2it(t^2-3)(t^6 - 33t^4 + 27t^2 - 3),
\end{aligned}$$

and

$$\begin{aligned}
(1+it)^9 + (1-it)^9 &= 2(9t^8 - 84t^6 + 126t^4 - 36t^2 + 1) \\
&= 2(3t^2-1)(3t^6 - 27t^4 + 33t^2 - 1).
\end{aligned}$$

Exercise 8.3

(b) By considering the equation $\tan 9\theta = 0$, or otherwise, obtain a cubic equation with integer coefficients whose roots are

$$\tan^2\left(\frac{\pi}{9}\right), \qquad \tan^2\left(\frac{2\pi}{9}\right) \qquad \text{and} \qquad \tan^2\left(\frac{4\pi}{9}\right).$$

(b) Take $\theta = \dfrac{\pi}{9}$. By (a), $t := \tan\left(\dfrac{\pi}{9}\right)$ satisfies

$$t(t^2-3)(t^6-33t^4+27t^2-3) = (\tan\pi)(3t^2-1)(3t^6-27t^4+33t^2-1) = 0.$$

Since $\tan\left(\dfrac{\pi}{9}\right) \neq 0, \pm\sqrt{3}$, $\tan^2\left(\dfrac{\pi}{9}\right)$ must be a root of the cubic equation

$$x^3 - 33x^2 + 27x - 3 = 0.$$

Similarly, we can show that $\tan^2\left(\dfrac{2\pi}{9}\right)$ and $\tan^2\left(\dfrac{4\pi}{9}\right)$ are roots of the equation $x^3 - 33x^2 + 27x - 3 = 0$ also.
So the required (monic) cubic equation is $x^3 - 33x^2 + 27x - 3 = 0$.

Exercise 8.3

(c) Deduce the value of

$$\tan\left(\frac{\pi}{9}\right)\tan\left(\frac{2\pi}{9}\right)\tan\left(\frac{4\pi}{9}\right).$$

(c) Let $\alpha_1, \alpha_2, \alpha_3$ be $\tan^2\left(\frac{\pi}{9}\right), \tan^2\left(\frac{2\pi}{9}\right), \tan^2\left(\frac{4\pi}{9}\right)$ respectively. Since $\alpha_1, \alpha_2, \alpha_3$ are the roots of $x^3 - 33x^2 + 27x - 3 = 0$, we have

$$\alpha_1\alpha_2\alpha_3 = -\frac{-3}{1} = 3.$$

Therefore

$$\tan\left(\frac{\pi}{9}\right)\tan\left(\frac{2\pi}{9}\right)\tan\left(\frac{4\pi}{9}\right) = \sqrt{3}$$

because $\tan\left(\frac{\pi}{9}\right), \tan\left(\frac{2\pi}{9}\right), \tan\left(\frac{4\pi}{9}\right) > 0$.

Exercise 8.3

(d) Show that

$$\tan^6\left(\frac{\pi}{9}\right)+\tan^6\left(\frac{2\pi}{9}\right)+\tan^6\left(\frac{4\pi}{9}\right)=33273.$$

(d) Note that

$$(\alpha_1+\alpha_2+\alpha_3)^3=\alpha_1^3+\alpha_2^3+\alpha_3^3+3(\alpha_1+\alpha_2+\alpha_3)(\alpha_1\alpha_2+\alpha_2\alpha_3+\alpha_3\alpha_1)-3\alpha_1\alpha_2\alpha_3.$$

Moreover, as roots of $x^3-33x^2+27x-3=0$, $\alpha_1,\alpha_2,\alpha_3$ satisfy

$$\begin{aligned}\alpha_1+\alpha_2+\alpha_3&=-\frac{-33}{1}=33,\\ \alpha_1\alpha_2+\alpha_2\alpha_3+\alpha_3\alpha_1&=\frac{27}{1}=27,\\ \alpha_1\alpha_2\alpha_3&=-\frac{-3}{1}=3.\end{aligned}$$

Therefore,

$$\begin{aligned}\tan^6\left(\frac{\pi}{9}\right)+\tan^6\left(\frac{2\pi}{9}\right)+\tan^6\left(\frac{4\pi}{9}\right)&=\alpha_1^3+\alpha_2^3+\alpha_3^3\\ &=33^3-3(33)(27)+3(3)=33273.\end{aligned}$$